Last but not least, as you add nodes, your blast radius decreases. For example, if you have five shards and experience a failover, only 20% of your data is affected. This means you can't write to this portion of the key space until the failover process completes (~15–30 seconds), but you can still read from the cluster, given you have a replica available. With cluster mode disabled, 100% of your data is affected, because a single node consists of your entire key space. You can read from the cluster but can't write until the DNS swap has completed.

You learned about the different deployment options for ElastiCache. There is one more thing we want to bring to your attention: AWS offers another in-memory database called MemoryDB, which you will learn about next.

11.2.5 MemoryDB: Redis with persistence

Even though ElastiCache for Redis supports snapshots and transaction logs for persistence, AWS recommends using ElastiCache as a secondary data store. AWS also provides an alternative called MemoryDB, a proprietary in-memory database with Redis compatibility and distributed transaction log. MemoryDB writes data to disk and reads from memory. By default, MemoryDB stores data among multiple data centers using a distributed transaction log. Therefore, AWS recommends using MemoryDB as a primary database. Keep in mind that data persistence comes with higher write latency—think milliseconds instead of microseconds.

So, MemoryDB is a good fit when a key-value store is all you need, but data persistency is essential, such as the following situations:

- *Shopping cart*—Stores the items a user intends to check out
- *Content management system (CMS)*—Stores blog posts and comments
- *Device management service*—Stores status information about IoT devices

MemoryDB has been generally available since August 2021. We haven't used MemoryDB for production workloads yet, but the approach sounds very promising to us.

Want to give MemoryDB a try? We have prepared a simple example for you. Create a CloudFormation stack by using the Management Console: http://mng.bz/jAJy.

While you wait for the stack to reach state `CREATE_COMPLETE`, let's have a look into the CloudFormation template necessary to spin up a MemoryDB cluster, shown here: